Name	Subject Class	Class	Candidate Number
	2BI		

ANGLO-CHINESE JUNIOR COLLEGE
Preliminary Examination 2017

BIOLOGY **9744/03**
Long Structured and Free-response Questions **21 Aug 2017**
PAPER 3 **2 hours**

Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your name, subject class, form class and index number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** questions in the spaces provided on the Question Paper.

Section B
Answer any **one** question in the spaces provided on the Writing Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show working or if you do not use appropriate units.

At the end of the examination, fasten your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

FOR EXAMINER'S USE	
1	
2	
3	
4 / 5	
TOTAL	**75**

This document consists of **17** printed pages.

For Examiner's Use

Section A

Answer **all** the questions in this section.

1 Dengue viruses (DENV) are responsible for millions of infections each year in tropical and subtropical areas of the world. According to the World Health Organization, dengue incidence has increased significantly over the past 50 years, turning this infection into the most important mosquito-borne disease in the world and a global health challenge. Fig. 1.1 shows the general global distribution of dengue fever in the year 2005.

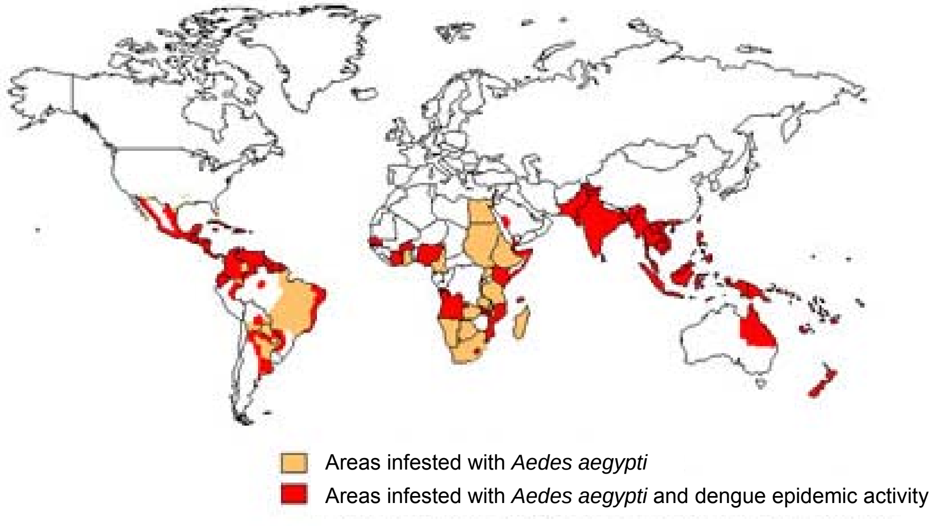


Fig. 1.1

(a) (i) Climate is one of the important factors that affects the distribution of dengue.

Predict and justify the expected distribution of dengue by the end of the 21st century.

____

____

____

____

____

____ [3]

(ii) Symptoms of dengue such as fever usually develop within 4 to 7 days after being bitten by an infected mosquito and is often associated with joint pain.

Explain how DENV may cause these symptoms.

____

____

____

____

____

____ [3]

For Examiner's Use

During an infection, DENV will elicit a primary humoral immune response which involves B cell activation as shown in Fig. 1.2.

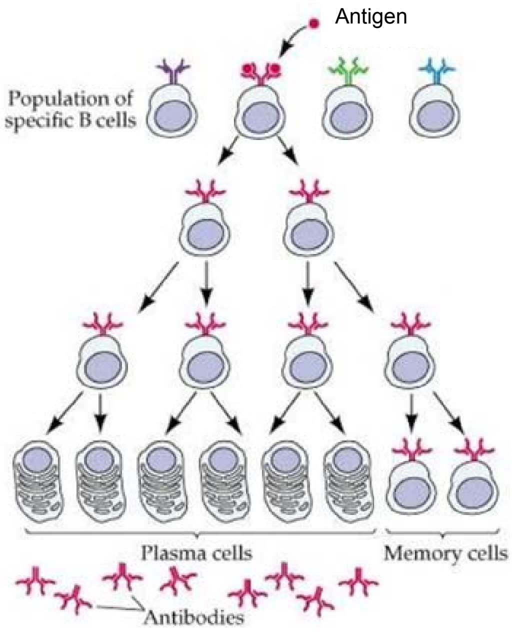


Fig. 1.2

(iii) With reference to Fig. 1.2 and your own knowledge, explain the significance of mitosis in B cell activation.

____

____

____

____

____

____ [3]

(iv) Memory B cells have long life spans because they do not actively undergo cell division. However, upon activation, a memory B cell can undergo many rounds of proliferation.

Suggest why activated memory B cells can undergo multiple rounds of cell division without dying.

____

____ [1]

(b) Bacteria has a natural defence system against viral infections involving RNA sequences known as clustered regularly interspaced short palindromic repeats (CRISPR).

When the bacteriophage infects a bacterial cell, the viral genome released into the bacterial cell is cleaved. Subsequently, a cleaved portion of the viral genome is integrated into the bacterial genome. The bacterial cell detects the phage DNA integrated within its genome and produces a type of RNA known as the CRISPR RNA. The CRISPR RNA contains a sequence that is complementary to that of the integrated phage DNA. When the next phage infects the same bacterial cell, the CRISPR RNA will bind to its target sequence in the viral genome and a nuclease is recruited to cut the phage DNA, disabling the invading phage.

The sequence of the CRISPR RNA can be edited and subsequently used to cut any DNA sequence at a precisely chosen site in eukaryotic cells. This gives rise to the possibility of correcting mutations associated with genetic disorders. Fig. 1.3 shows how the sequence of a defective allele can be replaced with the sequence of a normal allele from a donor DNA via homologous recombination using the CRISPR technology.

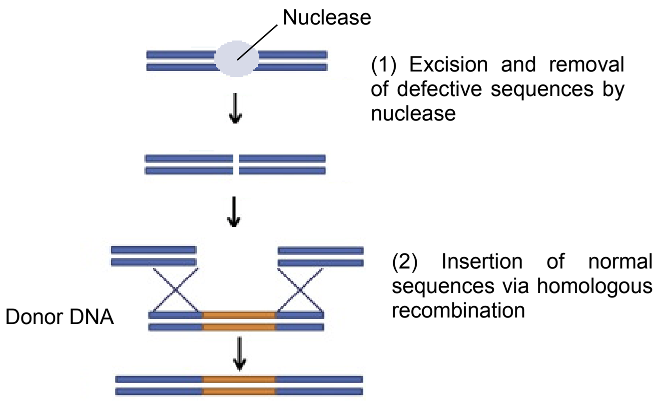


Fig. 1.3

(i) The specificity of CRISPR-mediated immunity in prokaryotes could be applied to eukaryotic cells to make precise changes in the genes of organisms.

Explain why the CRISPR RNA in prokaryotes can also be used in eukaryotic cells.

____

____

____

____ [2]

For Examiner's Use

(ii) Another technology that can be used to treat genetic disorders is gene therapy. Similar to the CRISPR technology, gene therapy involves the introduction of normal alleles into patients. However, it does not involve the excision and removal of defective alleles. Hence, it can only be used to treat recessive genetic disorder.

With reference to Fig. 1.3, explain the advantages of using CRISPR technology to treat genetic disorders over gene therapy.

____

____

____

____

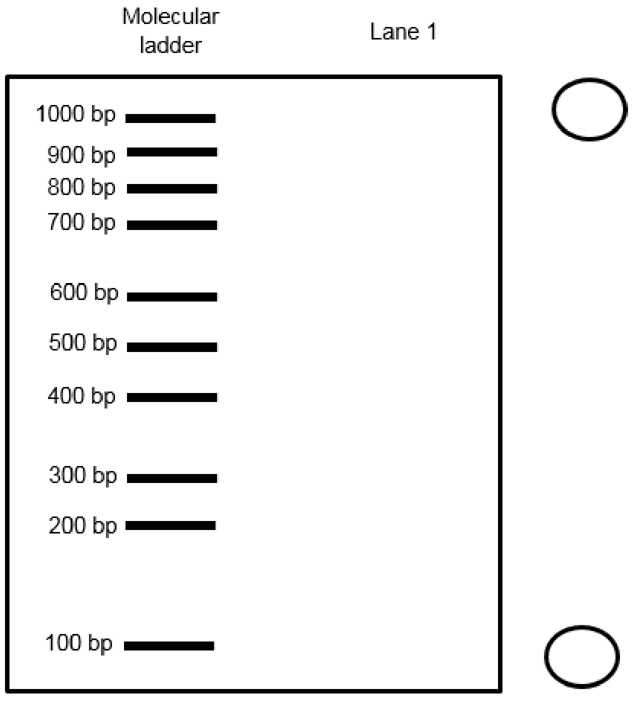
____

____ [3]

Polymerase Chain Reaction was used to amplify a gene of 1000 bp. Fig. 1.4 shows the resultant DNA fragment produced. A CRISPR RNA is designed to target a specific sequence found within the gene. The sites of excision by the nuclease are indicated by the arrows. Gel electrophoresis can be used to verify if the target sequence is cut at the precise sites by the nuclease.

Fig. 1.4

(iii) On the electrophoregram below, draw the expected results after the cut fragments are separated if there is specific binding of CRISPR RNA to target sequence in Lane 1. Indicate the charge of the electrodes clearly in the circles on the side of the electrophoregram.



[2]

ori

For Examiner's Use

Fig. 1.5 shows an electrophoregram obtained by a student following excision of the same gene by a nuclease.

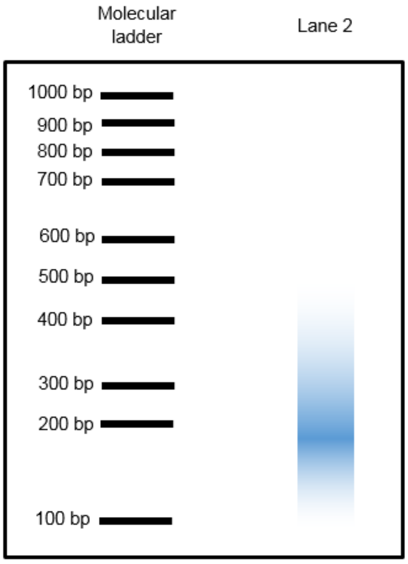


Fig. 1.5

(iv) Suggest how the smear is obtained in Fig. 1.5.

____

____ [1]

(v) Geneticists attempted to edit the gene responsible for β-thalassaemia, a potentially fatal blood disorder, in human embryos using the CRISPR technology. However, the experiment was terminated prematurely due to a number of unintended mutations and numerous ethical controversies.

Discuss the ethical implications of the application of CRISPR technology on genes in embryos.

____

____

____

____ [2]

For Examiner's Use

(c) One of the three tenets of the cell theory states that all living organisms are composed of one or more cells. Non-cellular life forms such as viruses challenge this tenet as they possess both living and non-living characteristics. The recent discovery of giant viruses, known as Mimiviruses, led scientists to rethink the origin of life and viral evolution. The features of the Mimivirus are as described.

1. It contains a double-stranded DNA genome of approximately 1.2 million base pairs which is significantly larger than the genomes of any other known virus and comparable to a cell.
2. It has some genes which show a high degree of homology to those in bacteria, while some show a high degree of homology to those in eukaryotes.
3. It contains a number of protein-coding genes showing high degree of homology to genes coding for products involved in translation, such as aminoacyl-tRNA synthetases and translation initiation factors. It also contains genes associated with metabolic pathways, DNA repair, and protein folding. However, it is still dependent on its host for translation.

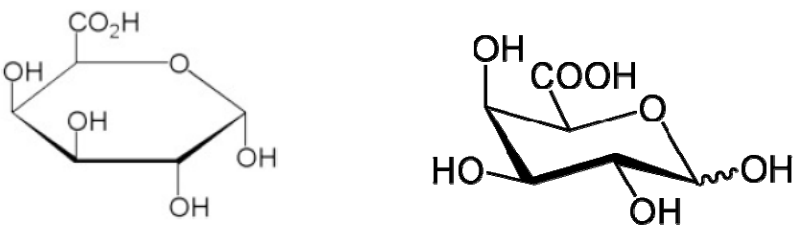
Suggest how Mimiviruses evolved and use the above statements to support your hypotheses.

[4]

[Total: 24]

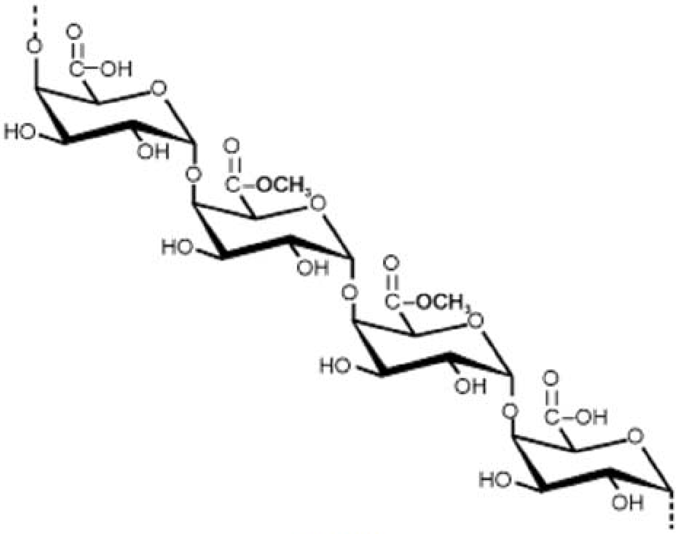
2 Prebiotics is defined as 'an ingredient that results in specific changes in the composition and activity of microbials in the gut thus improving host health'. Prebiotics are non-digestible oligosaccharides that can be selectively fermented by gut bacteria. Currently, most prebiotics are derived from the hydrolysis of simple polysaccharides. Therefore, to develop novel prebiotics, an alternative resource of polysaccharides that supplies oligosaccharides with more diverse and complex structures is required. One of these polysaccharides is pectin.

Fig. 2.1 shows the schematic diagram and natural occurring configuration of D-galacturonic acid, the monomer of pectin. The structure of pectin, a polysaccharide is shown in Fig. 2.2.



Schematic diagram — Natural occuring configuration

Fig. 2.1



Pectin

Fig. 2.2

For Examiner's Use

(a) With reference to Fig. 2.1 and 2.2, state two structural differences between cellulose and pectin.

....

....

....

.... [2]

Rhamnogalacturonan I (RG I) is a type of pectin present in the plant cell wall. The schematic diagram of RG I found in plant cell walls is shown in Fig. 2.3. 'Ac' represents acetyl groups added to RG I.

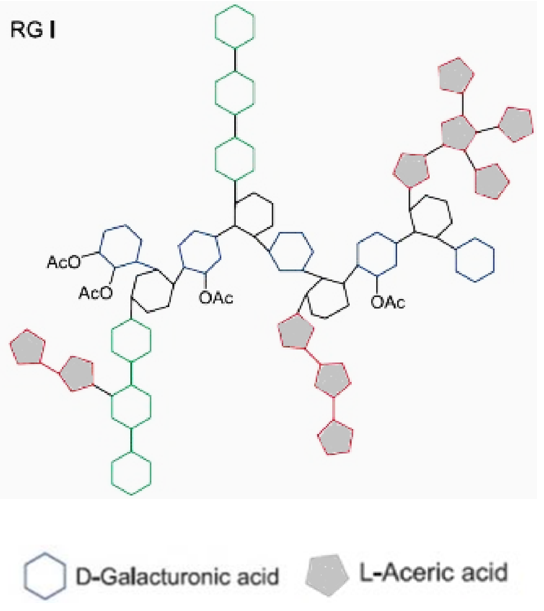


Fig. 2.3

(b) With reference to Fig. 2.3 and your knowledge of carbohydrates, suggest how the structure of monosaccharides allow for complexity in the pectin structure.

....

....

....

.... [2]

When pectin is consumed by humans, it is digested into oligosaccharides. These oligosaccharides are used by gut bacteria in the process of anaerobic respiration.

(c) Explain why a small yield of ATP can still be achieved with anaerobic respiration.

____

____

____

____

____

____ [3]

In the presence of oxygen, both glucose and triglycerides can be oxidised to yield large amounts of ATP. Fig. 2.4 shows some steps involved in respiration of triglycerides. Respiration of triglycerides involves the hydrolysis of triglycerides into fatty acid chains which are then broken down into acetyl CoA molecules. These acetyl CoA molecules will then enter the Krebs cycle.

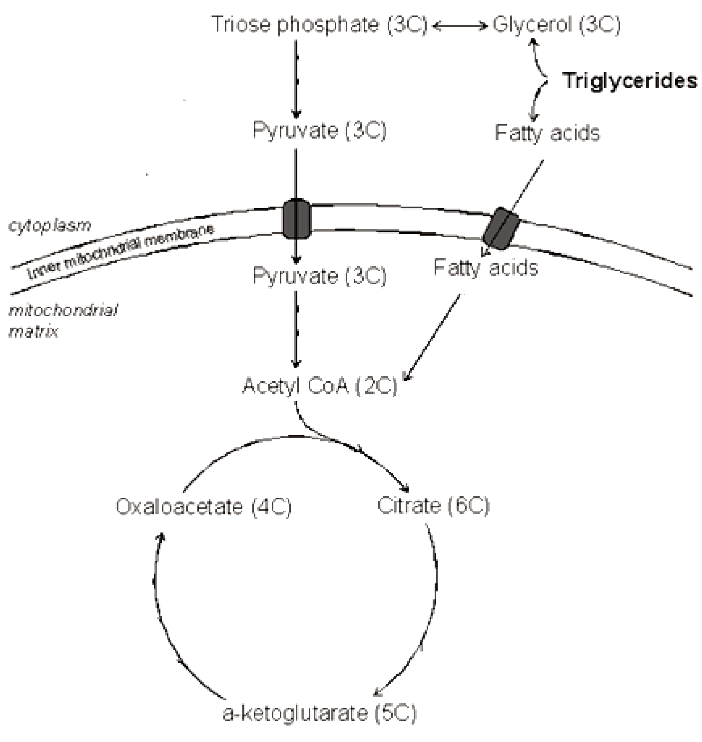


Fig. 2.4

(d) (i) Using the information from Fig. 2.4, calculate the total ATP yield from the complete oxidation of a 14-carbon fatty acid chain. Each NAD yields 2.5 molecules of ATP and each FAD yields 1.5 molecules of ATP. Show your calculation and answer in the space provided.

[3]

(ii) Explain why fatty acid chains can be completely oxidised only under aerobic conditions.

____

____

____

____

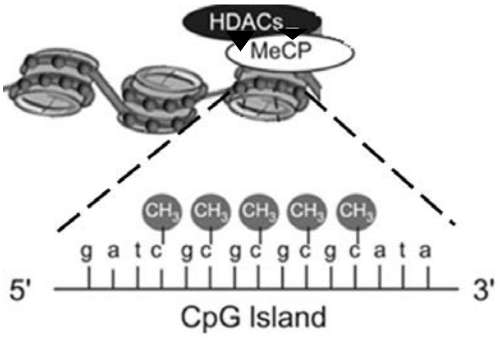
____

____ [3]

[Total: 13]

For Examiner's Use

3 CpG islands are regions of DNA where a cytosine nucleotide is followed by a guanine nucleotide in the linear 5' to 3' sequence. CpG islands are typically 300 to 3000 base pairs in length. These CpG islands have been found to be in or near approximately 40% of promoters of mammalian genes. Humans have a higher percentage of promoters with high CpG content.



HDAC: Histone Deacetylase
MeCP: Methyl-CpG-binding protein

Fig. 3.1

Commented [WU1]: Make the protein binding more complementary

(a) (i) With reference to the information provided, explain the significance of the presence of CpG islands in many of the genes in the mammalian genome.

____

____

____

____

____

____

____

____ [4]

McGhee and Ginder conducted an experiment that examined the effects of inhibiting methylation on gene expression. The experiment was performed using 5-azacytidine in mouse cells. 5-azacytidine is one of many chemical analogs that are structurally similar to the nucleoside cytidine from which cytosine is formed. When these analogs are integrated into growing DNA strands, some, including 5-azacytidine, severely inhibit the action of the DNA methyltransferase enzymes that normally methylate DNA. Interestingly, other analogs, like Ara-C, do not negatively impact methylation.

Scientists hypothesized that if they inhibited methylation by flooding cellular DNA with 5-azacytidine, then they could compare cells before and after treatment to see what impact the loss of methylation had on gene expression. The amount of methylation measured in each treatment is relative to the control. The results are shown in Table 3.1.

Table 3.1

Chemical Added	Number of Differentiated Cells	Amount of Methylation Measured
cytidine (control)	0	100%
Ara-C	0	127%
5-azacytidine	22141	33%

(ii) Explain what the experimental results show about the effects of methylation on gene expression.

____

____

____

____

____

____

____

____ [4]

(iii) Explain how confidence in the experimental results could be increased.

____

____ [1]

For Examiner's Use

(b) In 2010, a 10 year old boy with a damaged trachea (windpipe) was given a trachea transplant. A donor trachea was obtained and enzymes were used to remove all the cells, leaving only the collagen as shown in Fig. 3.2.

Fig. 3.2

Some bone marrow was removed from the boys' pelvis and about 2.5 million stem cells were isolated from this bone marrow. These stem cells were then treated with chemicals to stimulate proliferation and injected into the collagen.

The collagen, with the injected stem cells, was then immediately used to replace damaged trachea in the boy. Over a period of time, a fully functioning trachea was formed.

Suggest how the properties of the bone marrow stem cells allow for the formation of a fully functioning trachea.

[4]

[Total: 13]

For Examiner's Use

Section B

Answer **one** question in this section.

Write your answers to this question on the separate writing paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

4 **(a)** Using two named examples, explain the importance of bonds to the function' of two different classes of biomolecules. [13]

(b) With reference to named examples, describe the range of roles performed by lipids in living organisms. [12]

[Total: 25]

5 **(a)** Outline the processes that result in genetic variation in nature and explain the significance of such processes. [13]

(b) "The endomembrane system is critical in the synthesis of proteins". Discuss. [12]

[Total: 25]

END OF PAPER